4/10/26 SQL Data warehouse from scratch

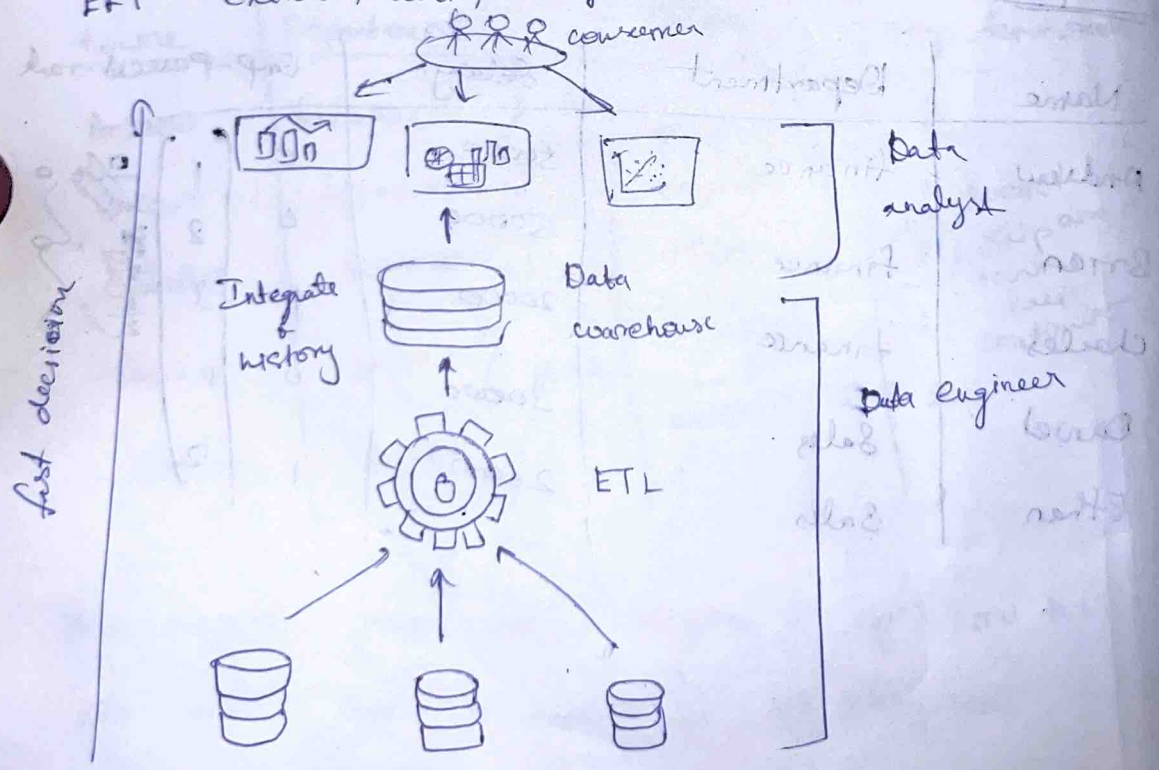
Data warehousing :- It is a process of collecting, integrating, storing and managing data from multiple sources in a central repository (Datawarehouse).

It enables organisation to organise large volume of current & historical data for efficient Querying, analysis & reporting.

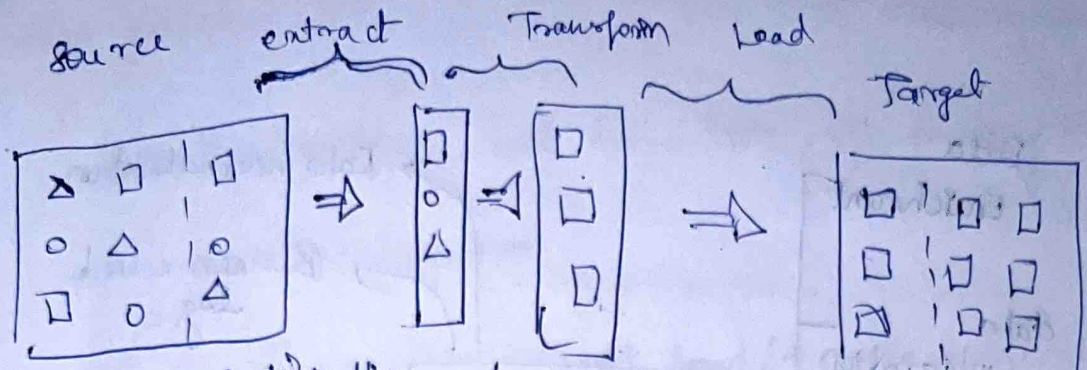
Datawarehouse :- subject oriented, integrated, time variant. (data not deleted / modified) - non volatile collection of data in support of management's decision - making process.

ETL - Extract, transform & load

ELT - extract, load, transform.



ETL.



identify the data we have to pull out from the source & not change anything (no manipulation).

Manipulation, transformation, data cleansing, normalization etc (reshaping to new format of what we need)

E - T - L

Extraction

Extract types

Extraction methods

Push Extraction

Pull extraction

Full Extraction

Incremental Extractions.

Extract techniques

manual data extraction

Database Querying

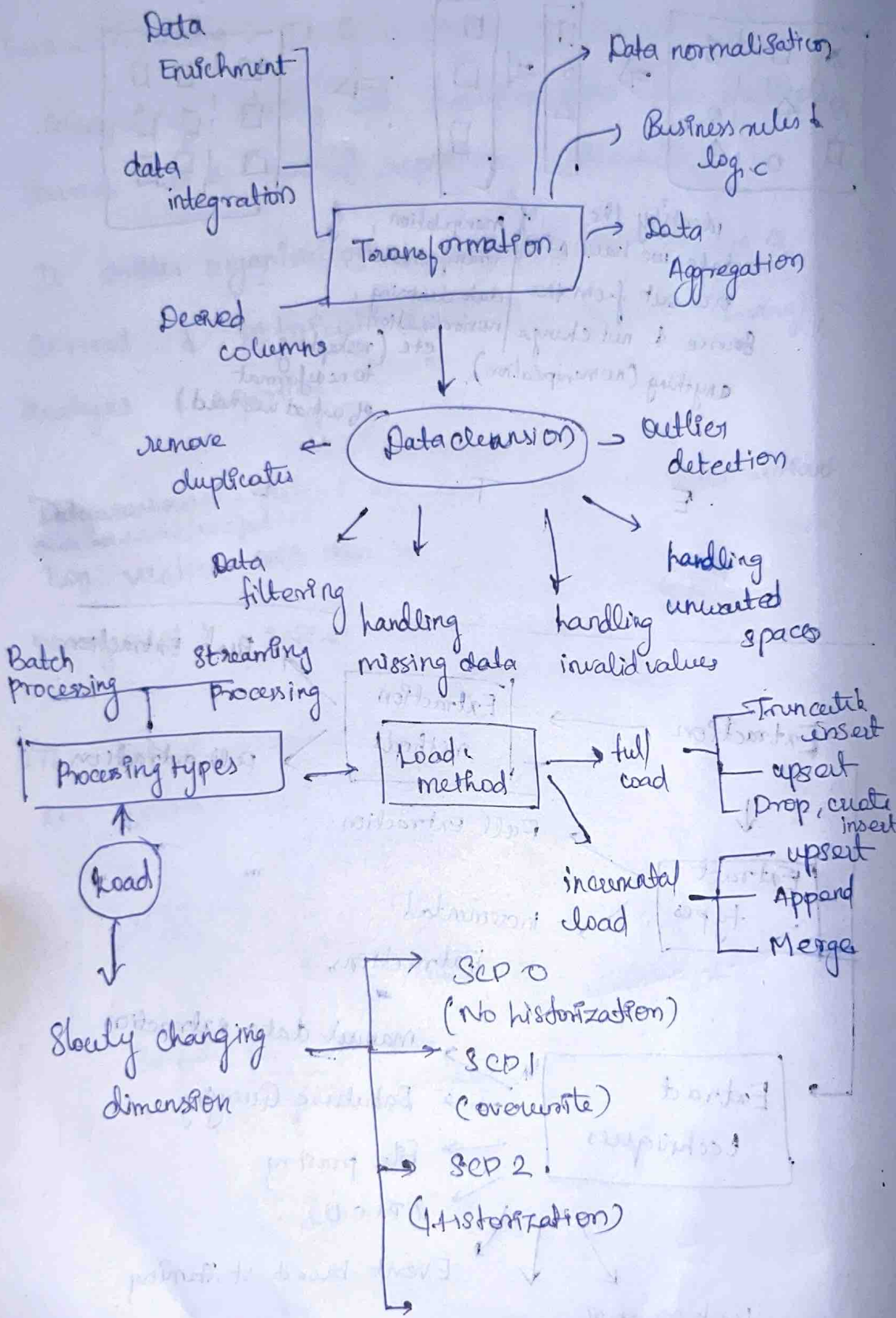
File parsing

API calls

Event based streaming

Webscraping

CDC


Data Enrichment
data integration
Transformation
Data normalisation
Business rules & logic
Data Aggregation
Desired columns
Data cleansing
remove duplicates
Outlier detection
Data filtering
handling missing data
handling invalid values
handling unwanted spaces
Batch Processing
Streaming Processing
Processing types
Load method
full load
Truncate insert
upsert
Drop, create insert
incremental load
upsert
Append
Merge
Load
Slowly changing dimension
SCD 0 (No historization)
SCD 1 (overwrite)
SCD 2 (Historization)

this project

→ pull extraction, full extraction, file parsing

→ All transformation

→ Batch processing, full load (truncate & insert), SCD1

1) Requirement analysis

Building a data warehouse ✓ ※

Develop a modern data warehouse using SQL server to consolidate Sales data, enabling analytical reporting & informed decision making

1) Datasources: import data from two source system (ERP & CRM) provided as csv file

2) Data Quality: cleanse & resolve data Quality issues prior to analysis

3) integration: combine both sources ~~System (ERP and CRM) provided~~ as files into a single, user friendly data model designed for analytical queries

4) Scope: focus on the lastest dataset only historization is not required

5) documentation: provide clear documentation of data model to support both business stakeholders & analytics team

BI: Analytic & reporting

develop SQL based analytics to deliver detailed insights into

→ customer behavior

→ product performance

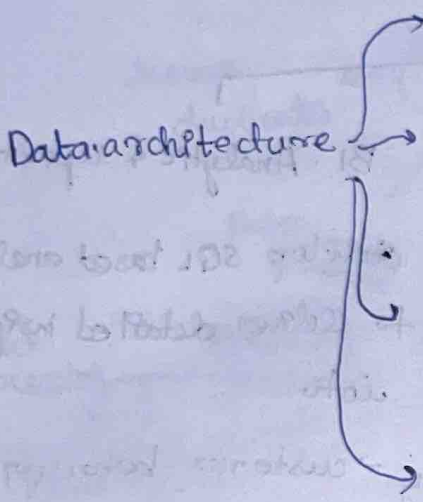
→ sales trends

These insights empower Stake holder with key business metrics, enabling strategic decision-ding

2) Design data architecture:

how data flows, integrates, and be accessed, ensure data warehouse is not only functioning and scalable, easy to maitain. (data architect)

→ Choose the right approach



1) Data warehouse is a centralized system that stores integrated, structured and generally cleaned data for reporting & analytical queries.

eg) online shopping company might combine order data, customer data, and payment data to calculate monthly revenue.

Source systems → ETL/ELT pipelines → Data warehouse
orders, customers, payments — Extract, transform, load — clean tables, facts, dimensions.

BI & reporting
Power BI, Tableau, SQL reports.

(Snowflake, Amazon redshift, google bigquery, Azure synapse Analytics)

2) Data Lake

A Data lake stores large volumes of data in many formats, often in its raw form. could be structured, semi structured & unstructured. data.

eg) CSV, JSON, parquet files, images, videos and clpd stream events

Data sources → Data Lake → Process when needed

Databases, APIs, Logs, Images — Amazon S3, Azure Data lake storage, gcp cloud storage — Python, Spark, Machine Learning, SQL engines.

3) Data Lakehouse

→ A Data lakehouse combines the flexible, scalable storage of a data lake with important warehouse capabilities such as structured tables, data governance & reliable concurrent analytics

→ Instead of maintaining completely separate systems for raw data and analytical data, an organisation can use a lakehouse architecture to support multiple workloads on a shared data foundation.

Data sources → cloud object storage → Lakehouse table & managment layer

S3 → ADLS, cloud storage — delta Lake, Apache Iceberg, Apache Hudi

Medallion architecture

Bronze: Raw ingested data

Silver: Cleaned, validated & deduplicated data

Gold: Business-ready & aggregated data

↓

SQL analytics — AI/ML

ETL pipelines — BI reporting

4) Data Mesh:

It is primarily an approach to organizing data ownership & delivery across an organization

eg) an bank with 4 depts

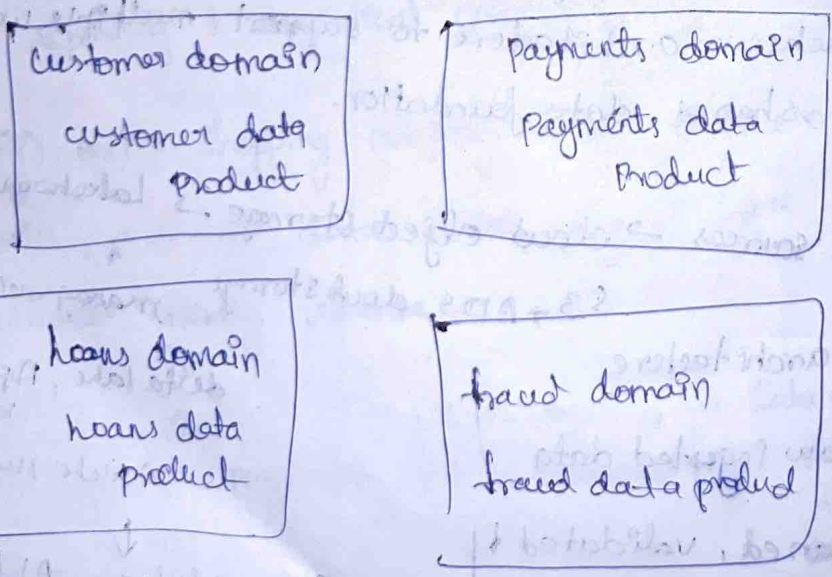
customer team
payments team
Loans team
fraud detection team

instead of single central data team being responsible for every dataset, each business domain owns its data & publishes well defined reusable data products

Shared standards & governance

Security, Quality, access policies, Metadata

↑↓



data mesh is about who owns the data, how it is shared and how teams work together. It does not require a specific storage Technology

Data mesh can use warehouse, lakes, lakehouses or combination

~~Datawarehouse~~ - ~~Snowflake/Redshift~~

	Bronze Layer	Silver Layer	Gold Layer
definition	Raw, unprocessed data as is from sources	Clean & standardized data (Intermediate Layer)	Business ready data
objective	Traceability & debugging	Prepare data for analysis	Provide data to be consumed for reporting & Analytics.
Object Type	Tables	Tables	Views
load method	Full Load (Truncate & insert)	full Load (Truncate & insert)	None
Data Transformation	None (as - is)	Data cleaning Data Standardization Data Normalization Derived columns Data enrichment	Data integration Data Aggregation Business logic & rules
Data modelling	None (as - is)	None (as-is)	→ Star schema → snowflake schema → Flat table
Target Audience	Data engineers	Data Analyst Data engineers	→ Aggregated objects - Data Analyst → Business users

Separation of concerns -

Separation of concerns means dividing a system into separate parts, where each part has its own specific responsibility.

(one component should focus on one main job instead of doing everything)

eg) A restaurant, where one person takes orders, another cooks and another handles billings. If everyone has a separate responsibility, the work becomes easier to manage, each team has different concern, but work together to achieve the same goal.

[if some rules change, only update that particular component without disturbing other]

SoC in data engineering

Imagine an ETL pipeline that loads sales data into Snowflake

Without SoC, a single large script might read files, validate data, clean records, calculate revenue, load tables and send email alerts

With SoC, we can divide the pipeline into dedicated steps

→ Extract - Read source data from file, APIs or databases

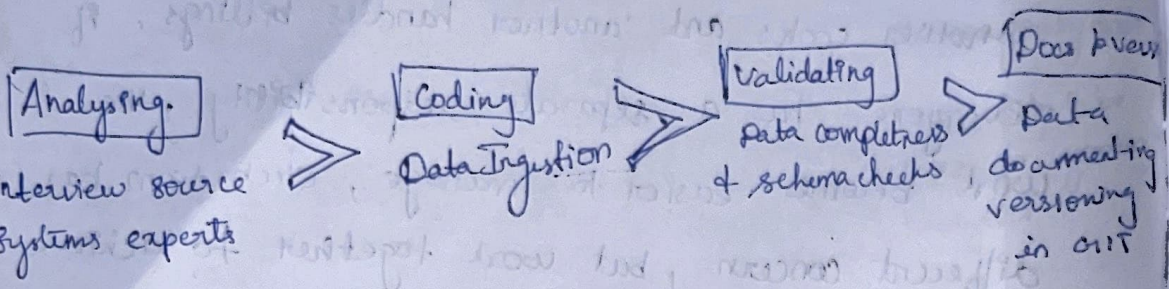
→ Validate - check required fields, nulls, duplicates and data types

→ Transform - clean records, apply business rules & calculate metrics

→ Load - write the transformed data into Snowflake tables

→ monitor - Record execution states, log errors and trigger alerts

Bronze layer



Business content & ownership

1) Who owns the data
2) what Business process it supports
3) System & data documentation
4) Data model & data catalog.

Architecture & Tech stack

1) How is data stored?
   (SQL server, Oracle, AWS, Azure)
2) What are the integration capabilities?
   (API, kafka, file Extract, Direct DB, ...)

Extract & load

Incremental vs full load

Data Scope & Historical needs

what is the expected size of all the extracts

Are there any data volume limitations?

How to avoid impacting the source system's performance

authentication & authorization

(tokens, SSH Keys, VPN / IP whitelisting)

Silver layer

Analysing → Coding → Validating → Doc & version

Analysing: Explore & understand the Data.

Coding: Data cleansing
- Check Quality of bronze
- write data transformation
- Insert into silver

Validating: Data correctness Checks

Doc & version: Data documenting versioning in GIT.

Gold layer

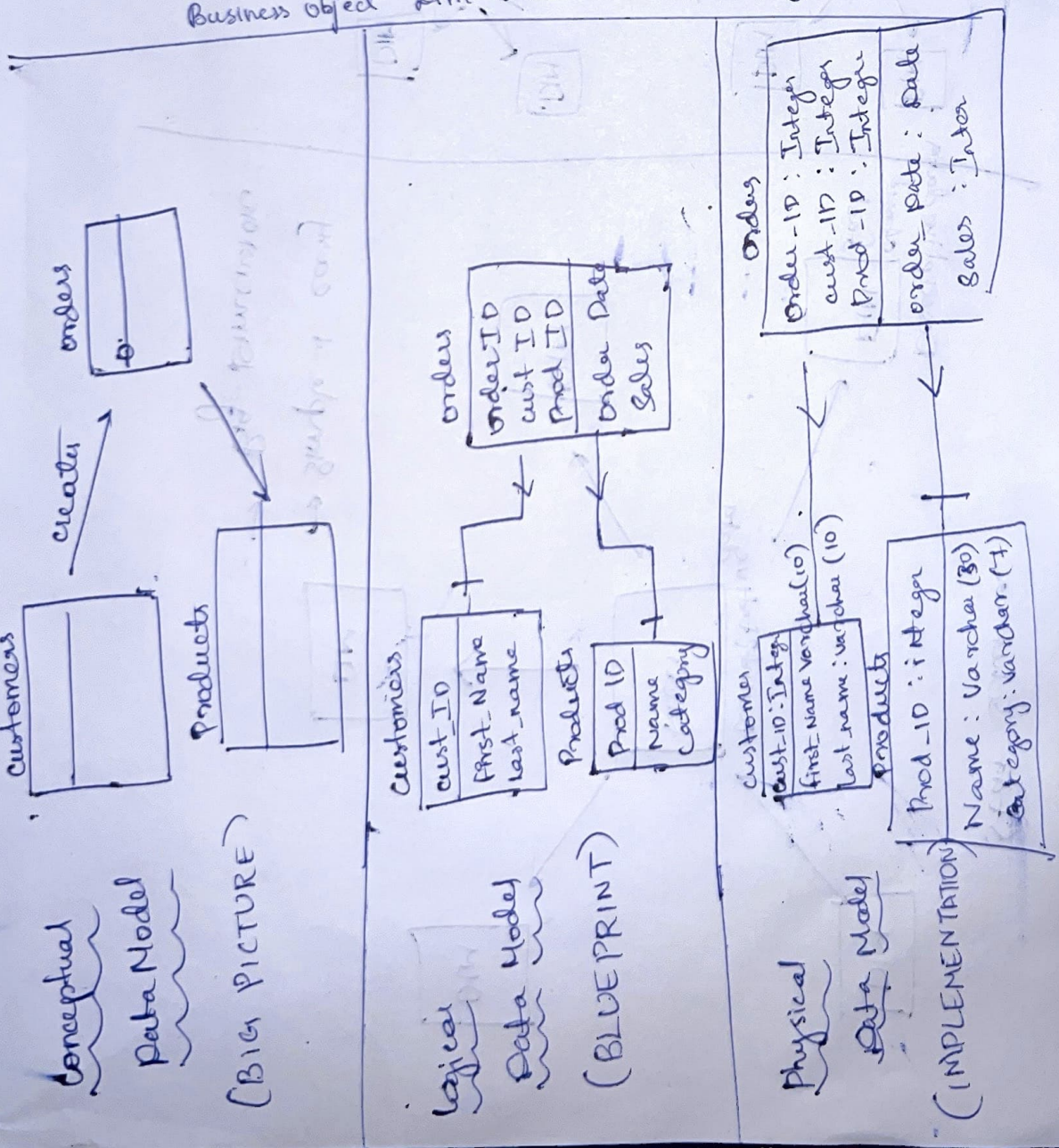
Analysing → Coding → Validation → Docs & version

Analysing: Explore & understand the Business objects

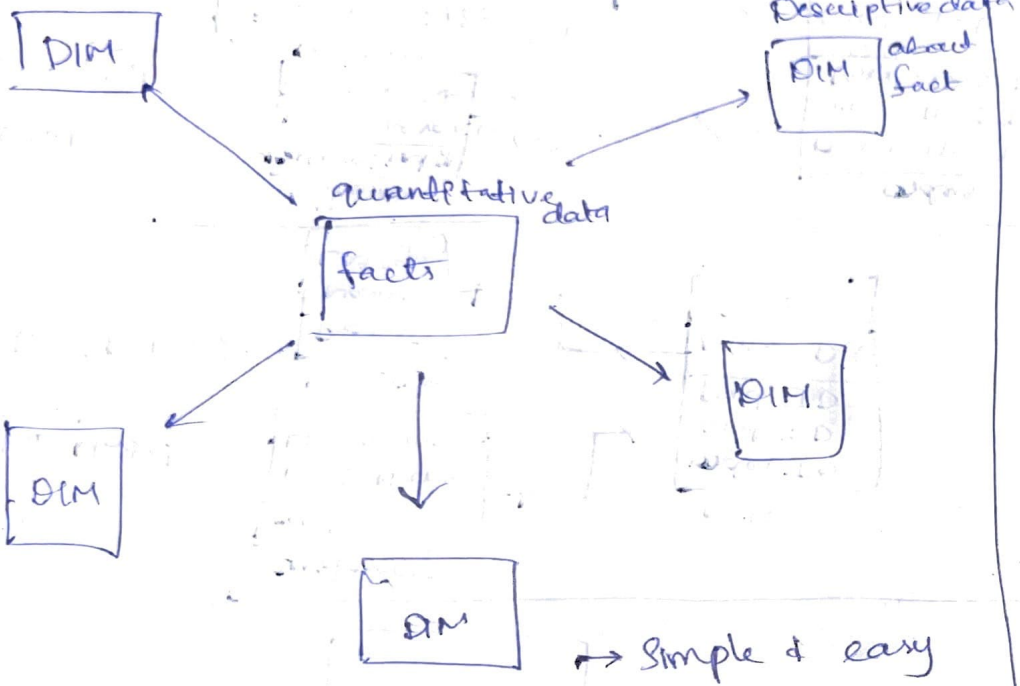
Coding: Data Integration
- Builds the Business object
- choose type Dim vs fact
- Rename to friendly name.

Validation: Data integration Checks

Docs & version: Documenting versioning in GIT



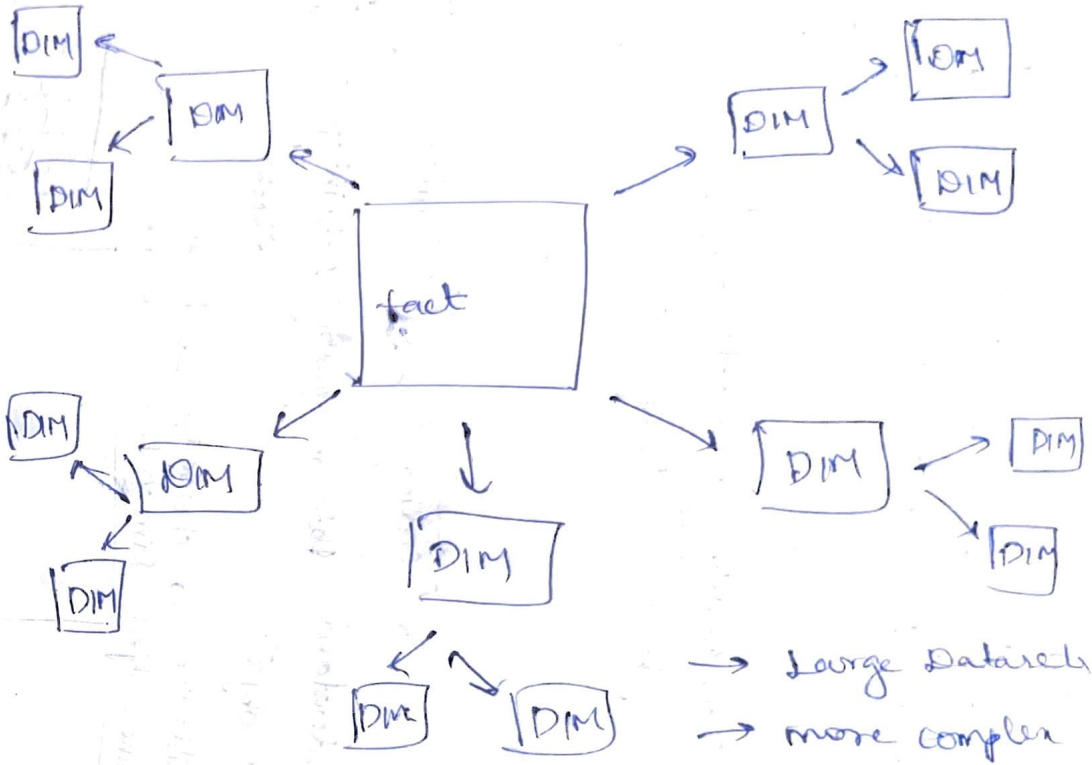
Star Schema



→ Simple & easy

→ Big dimensions

Snowflake Schema



→ Large Datasets

→ more complex

Dimension : Descriptive Information that give context to your data. who? what? where?

fact : Quantitive information that represents events how much? how many?